

N.S. LABORATORY SDN. BHD.

COSAMOX, 500mg/g, Powder for use in drinking water

Active Ingredient:

Each gram contains

Amoxicillin trihydrate ----- 574.1mg

(equivalent to Amoxicillin 500.0mg)

Product Description:

White or almost white powder which can be dissolved in animal drinking water.

Dosage form:

Powder for use in drinking water

Pharmacodynamics:

Amoxicillin is a time-dependent bactericidal antibiotic belonging to the semisynthetic penicillin group which acts by inhibiting the synthesis of bacterial cell walls during bacterial replication. It has a broad spectrum of activity against Gram positive and Gram negative bacteria, and owes its activity to the inhibition of the development of the peptidoglycan network structure in the bacterial cell wall.

Pharmacokinetics:

Amoxicillin is well absorbed following oral administration and it is stable in the presence of gastric acids. Excretion of amoxicillin is mainly in the unchanged form via the kidneys to give high concentration in renal tissue and urine. Amoxicillin is well distributed in body fluids. Studies in birds have indicated that amoxicillin is distributed and eliminated more rapidly than in mammals. Biotransformation appeared a more important route of elimination in birds than in mammals.

Indication:

Treatment of infections in chickens, turkeys and ducks caused by bacteria susceptible to amoxicillin.

Pigs: For the treatment of pasteurellosis caused by *Pasteurella multocida* susceptible to amoxicillin.

Mode of Administration:

Oral administration via drinking water.

Recommended Dosage:

Prepare the solution with fresh potable water immediately before use. Any medicated water which is not consumed within 24 hours should be discarded. Chickens- The recommended dosage is 15mg amoxicillin trihydrate per kg bodyweight per day (corresponding to 27mg product/kg bodyweight/day) for 3 days; in severe cases for 5 days.

Ducks- The recommended dosage is 20mg amoxicillin trihydrate per kg bodyweight per day (corresponding to 35mg product/kg bodyweight/day) for 3 consecutive days.

Turkeys- The recommended dosage is 15-20mg amoxicillin trihydrate per kg bodyweight per day (corresponding to 27-35mg product/kg bodyweight/day) for 3 days; in severe cases for 5 days.

Pigs- The recommended dosage is 20mg amoxicillin trihydrate per kg bodyweight (corresponding to 35 mg product/kg bodyweight) daily. The dose should be divided and administered at approximately 12 hourly intervals for up to 5 days.

The following formula may be used to calculate the required concentration of product:

$$\frac{\begin{matrix} x \text{ mg product per kg} \\ \text{bodyweight per day} \end{matrix} \times \begin{matrix} \text{mean bodyweight (kg)} \\ \text{of animals to be treated} \end{matrix}}{\begin{matrix} \text{mean daily water consumption (l) per animal} \end{matrix}} = \begin{matrix} x \text{ mg product per} \\ \text{litre drinking water} \end{matrix}$$

In order to ensure consumption of the medicated water, animals should not have access to other water supplies whilst being treated. Water supply system should be cleaned appropriately after the end of the medication period to avoid the intake of sub-therapeutic amounts of the active substance.

Contraindications:

Do not use in rabbits, hamsters, gerbils and guinea pigs, or to birds producing eggs intended for human consumption.

Not effective against beta-lactamase producing organisms.

Do not use in known cases of hypersensitivity to penicillin or other substances of the beta-lactam group or to any of the excipients.

Special Warnings for each Target Species:

Pigs: The uptake of medication by animals can be altered as a consequence of illness. In case of insufficient uptake of water, animals should be treated parenterally.

Special Precautions for Use in Animals:

Not effective against beta- lactamase producing organisms.

Official, national and regional antimicrobial policies should be taken into account when the product is used. Use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria. Inappropriate use of the product may increase the prevalence of bacteria resistant to the amoxicillin and may decrease the effectiveness of the treatment.

Special Precautions to be taken by the Person Administering the Veterinary Medicinal Product to Animals:

Penicillins and cephalosporins may cause hypersensitivity (allergy) following injection, inhalation, ingestion or skin contact. Hypersensitivity to penicillins may lead to cross reactions to cephalosporins and vice versa. Allergic reactions to these substances may occasionally be serious.

Do not handle this product if you know you are sensitised or if you have been advised not to work with such preparations.

Handle this product with great care to avoid exposure, taking all recommended precautions.

If you develop symptoms following exposure such as a skin rash, you should seek medical advice and show the doctor this warning. Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention.

Avoid inhalation of dust by wearing mask.

Wear gloves during preparation and administration of medicated water.

Wash any exposed skin after handling the product or medicated water.

Wash hands after use.

Interactions with other Medicaments:

Amoxicillin exerts its bactericidal action by inhibition of bacterial cell wall synthesis during multiplication. It is therefore in principle not compatible with bacteriostatic antibiotics (e.g. tetracyclines, macrolides and sulphonamides) which inhibit multiplication. Synergism occurs with β -lactam antibiotics and aminoglycosides.

Usage during Pregnancy and Lactation:

The safety of the veterinary medicinal product has not been established during pregnancy and lactation. Laboratory studies in rats and rabbits have not produced any evidence of teratogenic, foetotoxic or maternotoxic effects. Use only according to the benefit/risk assessment by the responsible veterinarian.

Adverse Effects:

Penicillins and cephalosporins may cause hypersensitivity following administration. Allergic reactions to these substances may occasionally be serious.

Overdose and Treatment:

In case of overdosing the treatment should be symptomatic. No specific antidote is available.

Withdrawal Periods (Meat and Offal):

Chickens- 1 day; Ducks- 9 days; Turkeys- 5 days; Pigs- 2 days

Not for use in birds producing or intended to produce eggs for human consumption.

Storage Condition:

Store in cool, dark place (< 30°C)

Shelf Life:

36 months

Shelf Life after Opening:

Discard any unused powder immediately after opening

Shelf Life after Dilution/Reconstitution:

24 hours

Packaging Available:

100g/packet, 100packets/carton, 10kg/carton;

500g/packet, 30 packets/carton, 15kg/carton

Manufacturer and Registration Holder:

N.S. Laboratory Sdn. Bhd.

54, Jalan S2C2, Green Technology Park, Seremban 2, 70300 Seremban, Negeri Sembilan, Malaysia.

Revised Date:

08/01/2023